

MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Data Cleaning & Transformation:

The original dataset contained **45,000 rows and 34 columns**, with no duplicate records. However, to practice and demonstrate data-cleaning techniques, I manually introduced data inconsistencies, including **2% duplicate rows, 1% missing values, 5% inconsistent department names, 2% inconsistent healthcare indicators, and 2% blood group values converted to lowercase**. After introducing these issues, appropriate data-cleaning techniques were applied, such as duplicate removal, missing-value handling, standardisation of department names, correction of healthcare indicators, and formatting blood group values consistently. Required transformations and calculated fields were then created to prepare the final dataset for KPI calculation and dashboard development.

Data Cleaning Techniques

Data Cleaning is the process of identifying and correcting inaccurate, incomplete, duplicate, or inconsistent data. It was used to improve data quality, maintain consistency, and ensure accurate KPI calculations and dashboard results.

1. **Duplicate Removal** – Removed duplicate records to avoid repeated data and incorrect KPI calculations.
2. **Missing Value Handling** – Identified and handled missing values to prevent errors during analysis.
3. **Department Name Standardisation** – Standardised inconsistent department names for accurate department-wise analysis.
4. **Healthcare Indicator Correction** – Corrected inconsistent values in healthcare indicators such as admission type and admission status.
5. **Blood Group Standardisation** – Converted lowercase blood group values into a consistent format for accurate filtering and grouping.
6. **Data Type Correction** – Converted dates, numerical values and categorical fields into appropriate data types.

Data Transformation Techniques

Data Transformation is the process of converting and organising cleaned data into a suitable format for analysis. It was used to create calculated fields, categorise data, integrate tables, and prepare meaningful information for dashboard visualisation.

1. **Date Transformation** – Converted admission and discharge dates into proper date formats for monthly and yearly analysis.
2. **Age Group Creation** – Created age groups to analyse patient demographics.
3. **Length of Stay Calculation** – Calculated the number of days between admission and discharge dates.
4. **Data Aggregation** – Summarised data by department, ward, disease category and month.
5. **Financial Transformation** – Prepared total amount, insurance-covered amount and patient-payable amount for financial analysis.

KPI Calculations:

A **Key Performance Indicator (KPI)** is a measurable value used to evaluate performance, efficiency and progress of an organisation or process. KPIs help decision-makers understand the current situation, identify problems and make better decisions based on data.

How KPIs Are Useful in Our Project

Our project is a Hospital Operations & Patient Analytics Dashboard. The purpose of our project is to analyse hospital and patient data and provide useful insights through Python, SQL and Tableau. KPIs are useful because they convert large amounts of hospital data into simple numbers that management can understand and monitor. In our hospital analytics project, KPIs monitor key healthcare operations and measure hospital performance. They provide a quick summary of key information through dashboard cards and help hospital management identify areas that need improvement.

KPI's Used in This Project:

1. **Total Admissions** – Measures the total number of patient admissions and helps understand hospital workload.
Formula: Total Admissions = COUNT(Admission_ID)
2. **Total Patients** – Shows the number of unique patients treated in the hospital.
Formula: Total Patients = COUNTD(Patient_ID)
3. **Total Discharges** – Measures the number of patients discharged and helps analyse patient flow.
Formula: Total Discharges = COUNTD(Admission_ID), where Discharge_Date is not null
4. **Occupancy Rate** – Shows the percentage of hospital beds occupied by patients and helps monitor bed availability.
Formula: Occupancy Rate (%) = (Number of Occupied Beds / Total Beds) × 100
6. **Readmission Rate** – Measures how frequently patients are admitted again and helps identify possible quality-of-care issues.
Formula: Readmission Rate (%) = (Number of Patients with More Than One Admission / Total Patients) × 100
7. **Average Length of Stay** – Shows the average number of days patients stay in the hospital and helps evaluate treatment efficiency and bed usage.
Formula: Average Length of Stay = SUM(Discharge_Date - Admission_Date) / Number of Discharged Patients
8. **Bed Utilisation Rate** – Measures how effectively hospital beds are being used.
Formula: Bed Utilisation Rate (%) = (Number of Patients / Total Bed Capacity) × 100
9. **Department Efficiency** – Helps compare department performance based on admissions, discharges, resources and operational activities.
Formula: Department Efficiency (%) = (Department Output / Department Resources) × 100
10. **Total Revenue** – Shows the total billing amount generated by the hospital.
Formula: Total Revenue = SUM(Patient_Payable_Amount + Insurance_Covered_Amount)
11. **Total Staff** – Measures the number of employees available for hospital operations.
Formula: Total Staff = COUNTD(Employee_ID)
12. **Total Wards** – Shows the total number of wards available in the hospital.
Formula: Total Wards = COUNTD(Ward_ID)
13. **Total Beds** – Measures the total bed capacity of the hospital.

Formula: Total Beds = SUM(Total_Beds)

14. Average Beds per Ward – Shows the average number of beds available in each ward and helps evaluate ward capacity.

Formula: Average Beds per Ward = Total Beds / Total Wards

Overall, these KPIs help hospital management monitor performance, improve resource utilisation, identify operational issues and make data-driven decisions.